WEB TECHNOLOGY LAB

EXP.NO: 4

DATE: 26/02/2025

AIM:

To create a simple web application that includes a form to accept a user's name and age, and then use a Servlet to process the submitted data. The Servlet will retrieve the form data, process it, and display the values back on the browser.

ALGORITHM:

1. Input: The user fills out a web form with their name and age and submits it.

2. Create the Web Form:

   - Create an HTML form with two input fields: one for the user's name and one for the user's age.

   - Add a submit button to the form.

   - Set the form's action attribute to the URL of the Servlet and the method to POST.

3. Create the Servlet Class:

   - Write a Servlet that processes the form data submitted by the user.

   - Override the doPost() method to handle form submissions.

   - Retrieve the name and age using request.getParameter().

4. Process the Data in the Servlet:

   - Extract the name and age values submitted from the form.

   - Validate the age input.

   - Prepare a response to display the user's name and age.

5. Send Response to the Browser:

   - Display message: 'Hello, [name]! You are [age] years old.'

6. Deploy the Servlet:

   - Compile the Servlet and deploy it in Apache Tomcat.

- Map the Servlet using annotations or web.xml.

7. Test the Application:

- Open browser, enter name and age, submit the form.

- Verify the displayed output.

PROGRAM:

1. HTML Form (index.html)

```
<!DOCTYPE html>
<html>
<head>
  <title>User Form</title>
</head>
<body>
  <h2>User Information Form</h2>
  <form action="UserServlet" method="post">
    Name: <input type="text" name="name" required><br><br>
    Age: <input type="number" name="age" required><br><br>
    <input type="submit" value="Submit">
  </form>
</body>
</html>
```

2. Servlet Program (UserServlet.java)

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/UserServlet")
public class UserServlet extends HttpServlet {

  protected void doPost(HttpServletRequest request,
                  HttpServletResponse response)
        throws ServletException, IOException {
```

```
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        String name = request.getParameter("name");
        String age = request.getParameter("age");

        out.println("<html><body>");
        out.println("<h2>Hello, " + name + "! You are " + age + " years old.</h2>");
        out.println("</body></html>");
    }
}
```

3. web.xml (If Annotation Not Used)

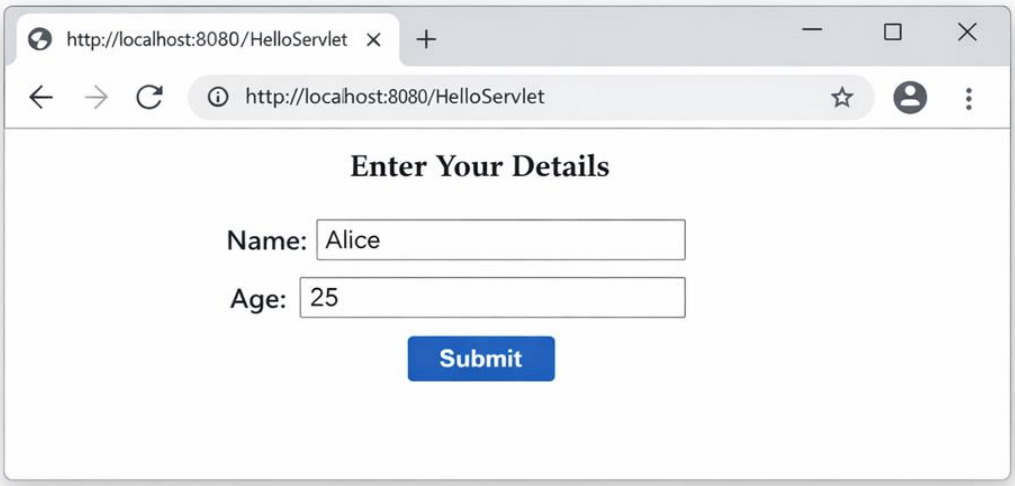
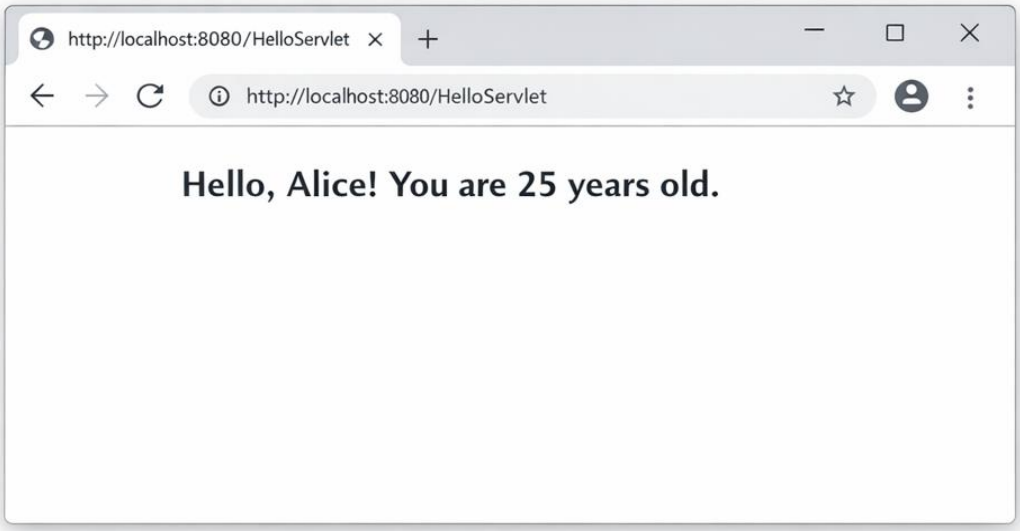
```
<web-app>
  <servlet>
    <servlet-name>UserServlet</servlet-name>
    <servlet-class>UserServlet</servlet-class>
  </servlet>

  <servlet-mapping>
    <servlet-name>UserServlet</servlet-name>
    <url-pattern>/UserServlet</url-pattern>
  </servlet-mapping>
</web-app>
```

RESULT:

A Servlet is successfully created to process the HTML form data and display it back on the browser.

Input Form



Output Display